Antigen processing and presentation

INIM 0006 2025

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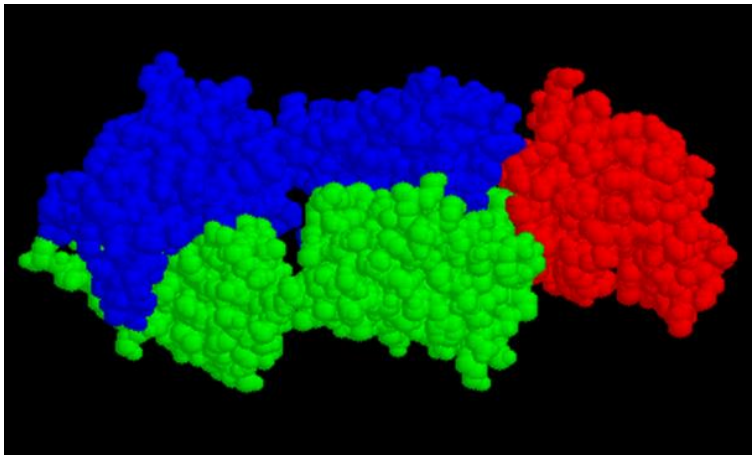
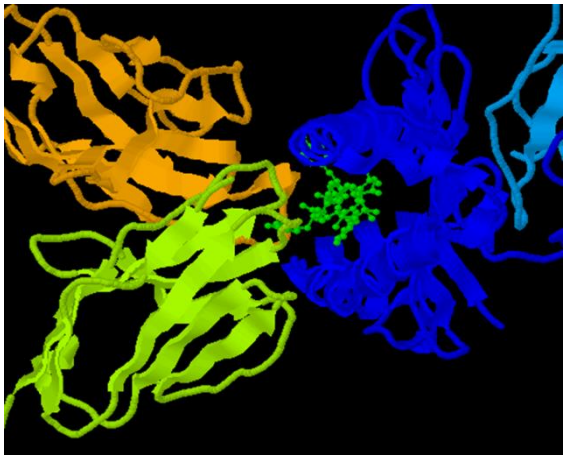
https://www.innate2adaptive.uk/

Intended learning outcomes

- To revise the fundamental differences between antigen recognition by T cells and B cells
- To revise the structure of Class I and Class II MHC molecules and their function in T cell antigen recognition
- To know the basic genetics of the MHC complex, and to appreciate the role of genetic polymorphism in immune recognition
- To understand the cell biology of the endogenous MHC Class I and the exogenous MHC Class II pathway of antigen processing
- To understand the role of the accessory molecules like invariant chain in regulating antigen processing
- To appreciate that microbes and cancer cells exhibit a diverse set of mechanisms to interfere with antigen processing and therefore reduce T cell recognition

Fundamental differences between T and B cell recognition





Chapter 6 of Janeway's immunobiology 10th Edition 2022

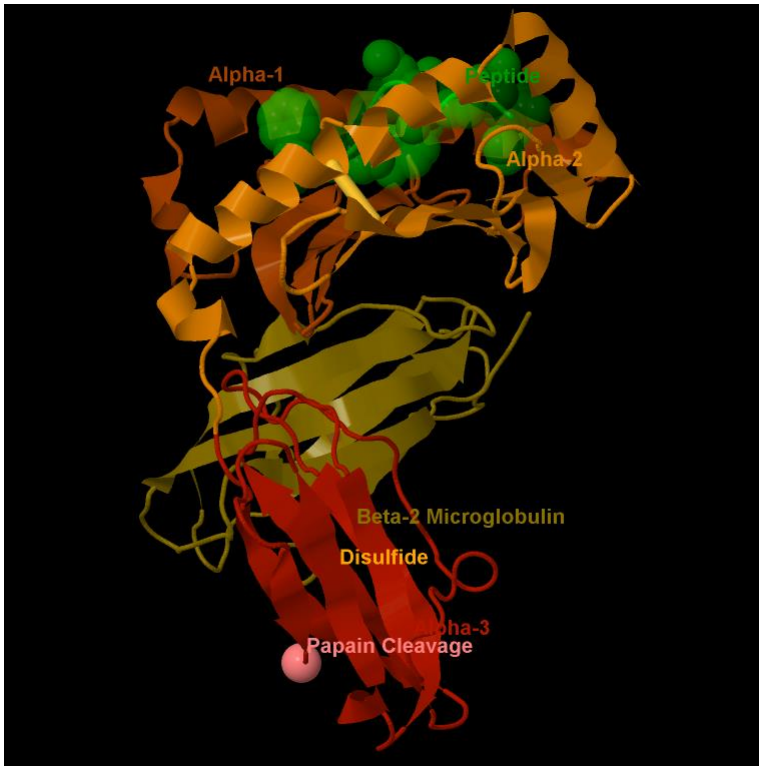
I The molecular basis of antigen recognition by T cells.

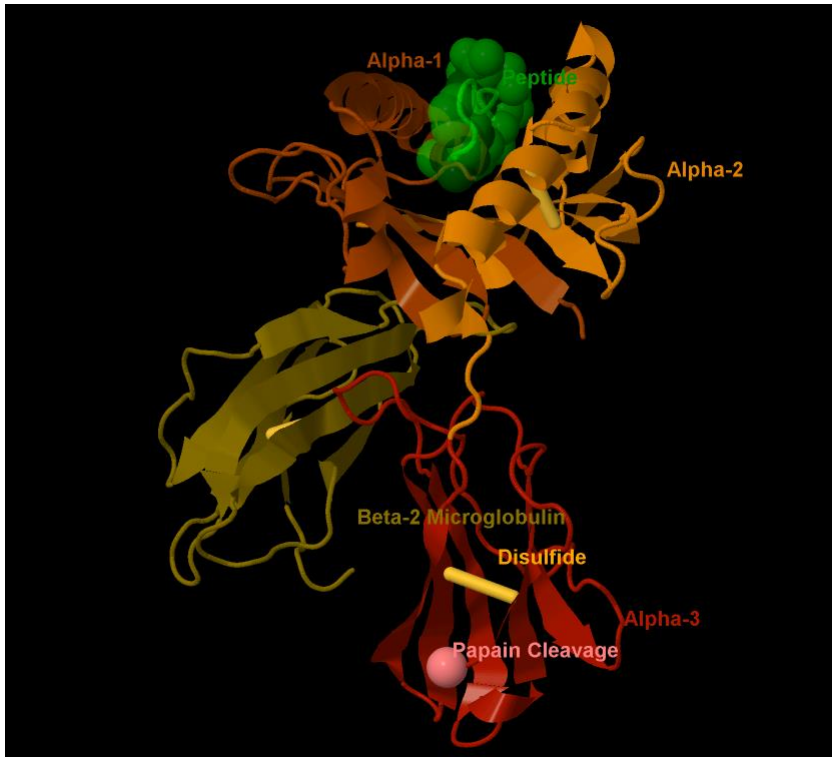
The major histocompatibility complex (MHC) molecules as antigen binding structures

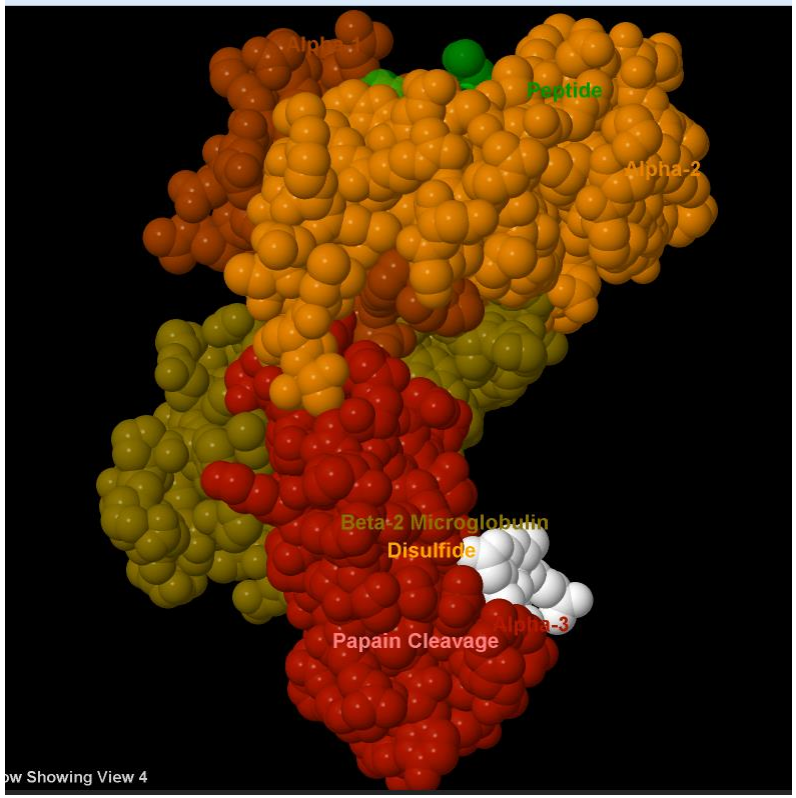
- Crystallography : peptide in the groove
- Role of peptide in stability
- MHC bound peptide motifs
- Epitope prediction and cancer
- Non-classical binding molecules

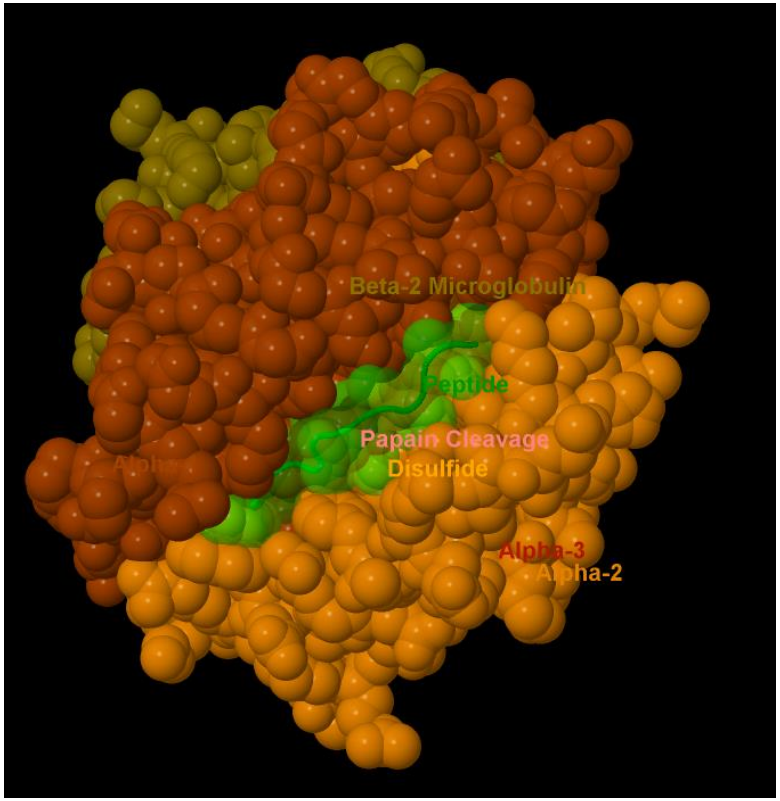
For structures see here :

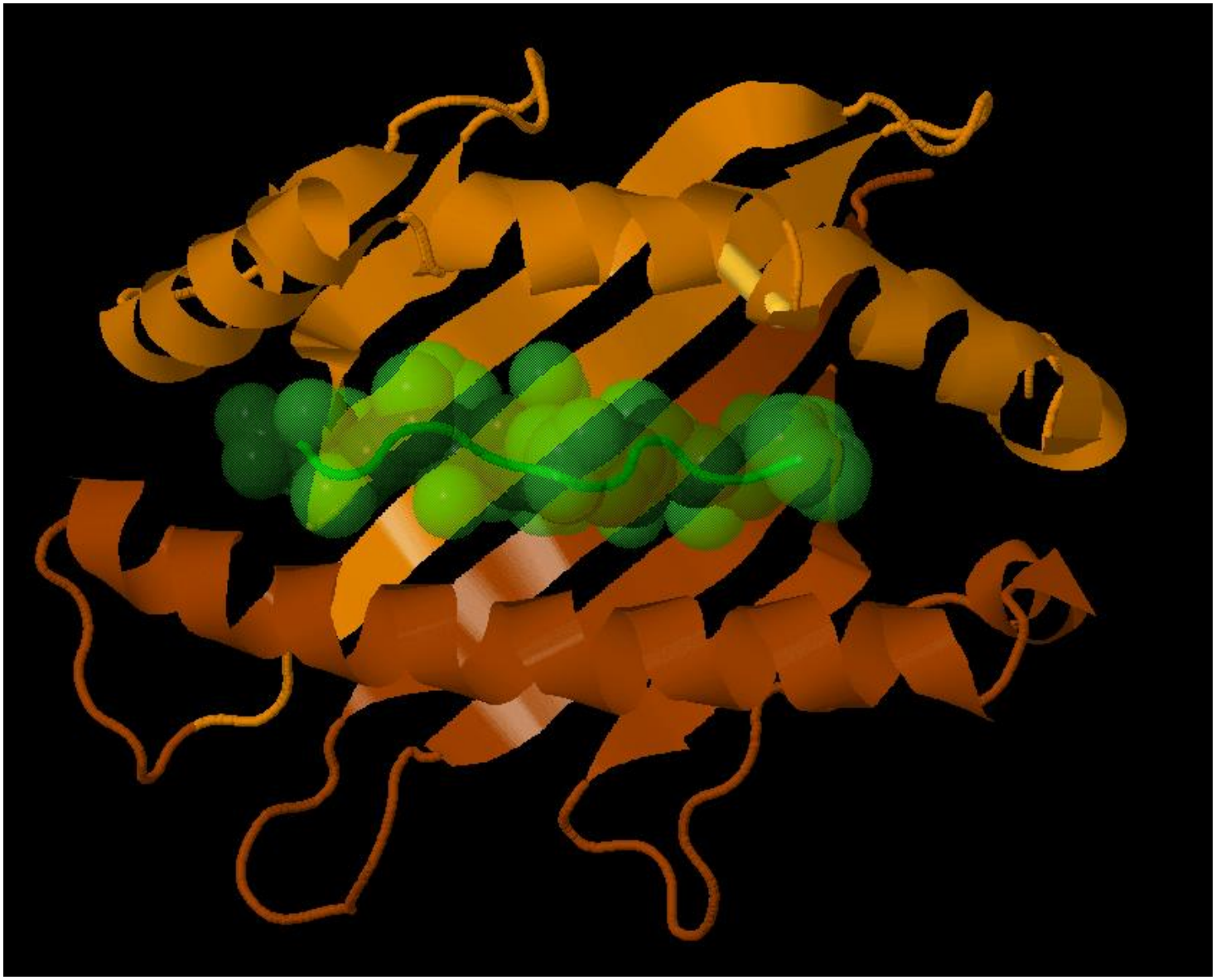
MHC Class I
MHC Class II

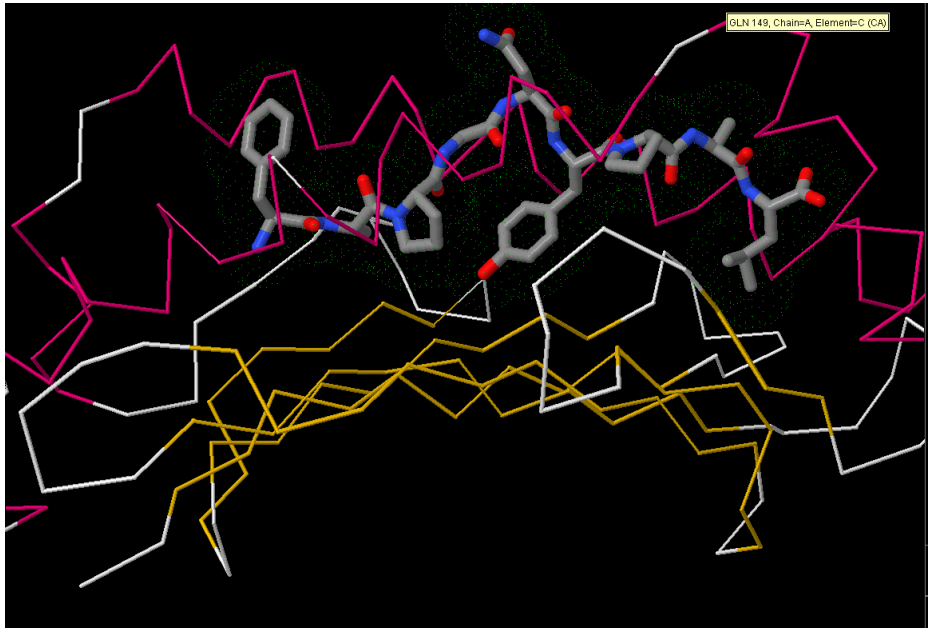

Alpha-1
Peptide
Alpha-2
Beta-2 Microglobulin
Disulfide
Papain Cleavage

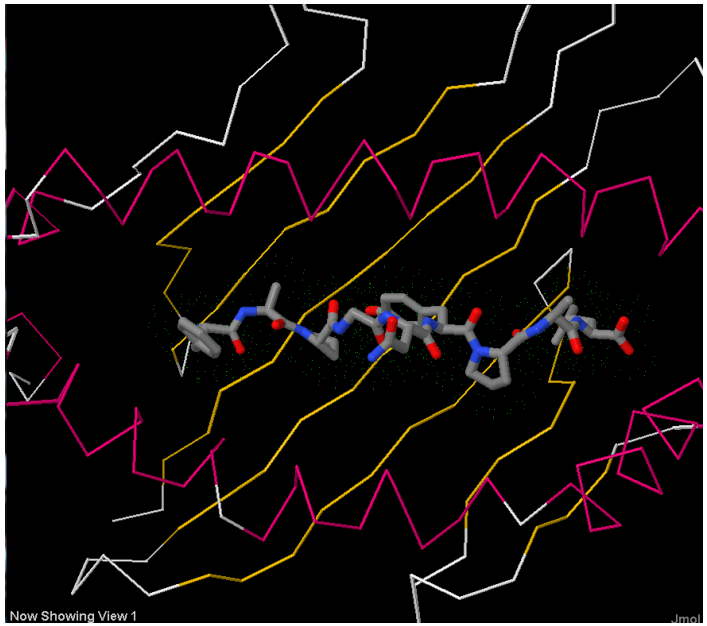

Alpha-1
Peptide
Alpha-2
Beta-2 Microglobulin
Disulfide
Alpha-3
Papain Cleavage

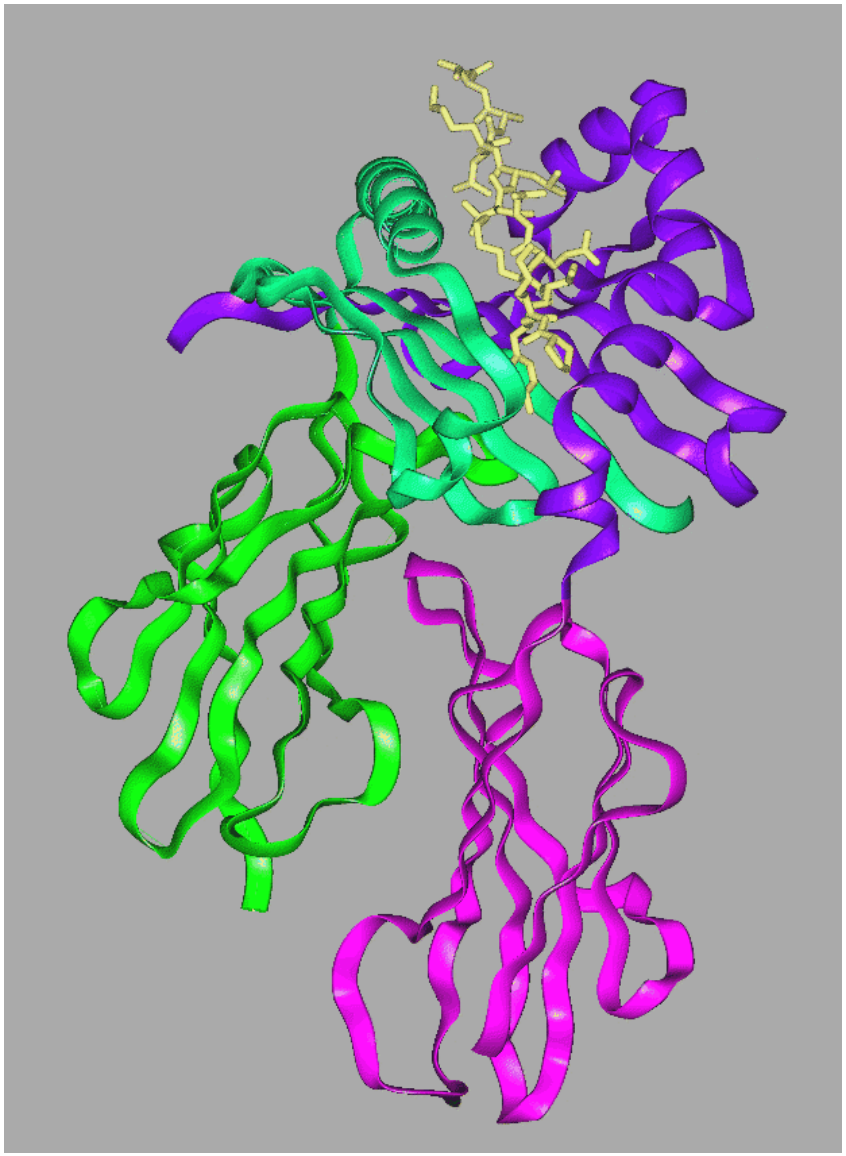

Peptide
Alpha-2
Beta-2 Microglobulin
Disulfide
Papain Cleavage
ow Showing View 4


Beta-2 Microglobulin
Peptide
Papain Cleavage
Disulfide
Alpha-3
Alpha-2




GLN 149, Chain=A, Element=C (CA)

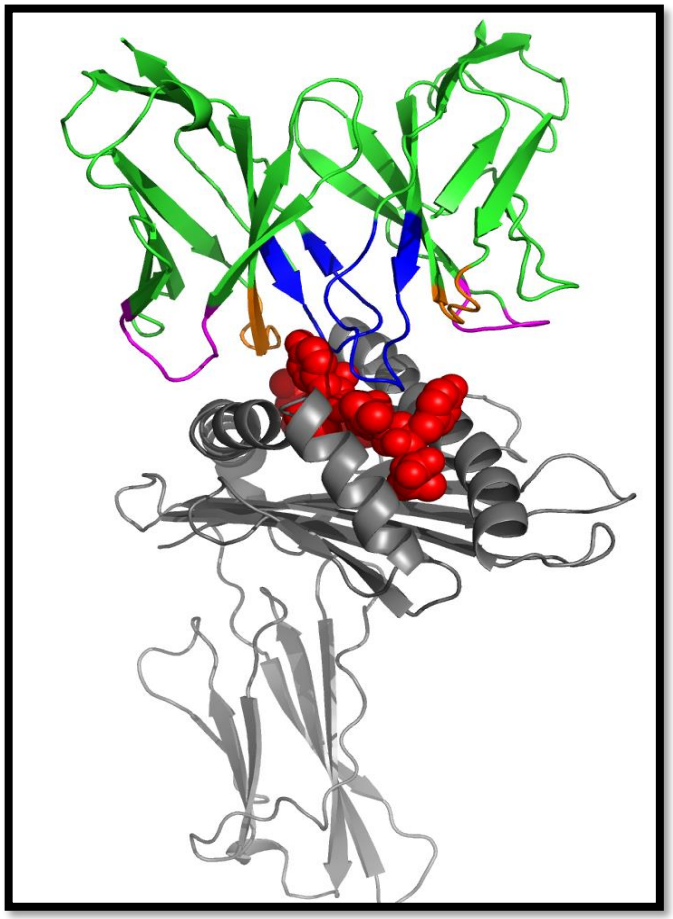

Now Showing View 1
Jmol



Class II MHC – note two chains and more open-ended binding groove.

The three (or four) body problem – binding the TCR

- An α and a β chain
- 3 hyper-variable regions generated by random recombination and insertion.
  - **CDR1** – known to form contacts with the MHC complex
  - **CDR2** – known to form contacts with the MHC complex
  - **CDR3** – believed to be the key determinant of TCR specificity for peptide



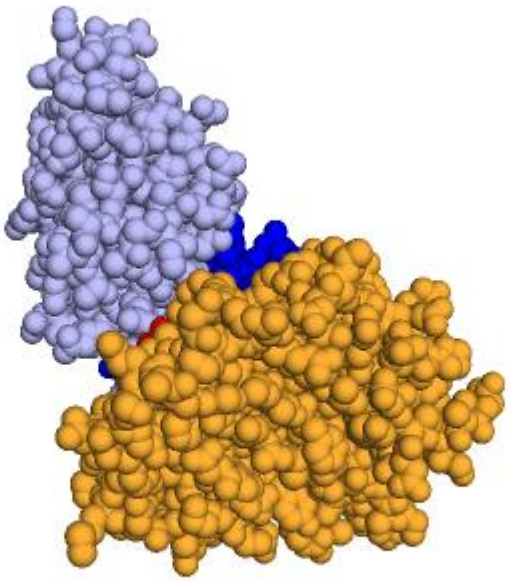
The T cell receptor

TCR alpha

TCR beta

MHC

Antigen



MHC:TCR:Peptide (1AO7) HLA A02 Tax

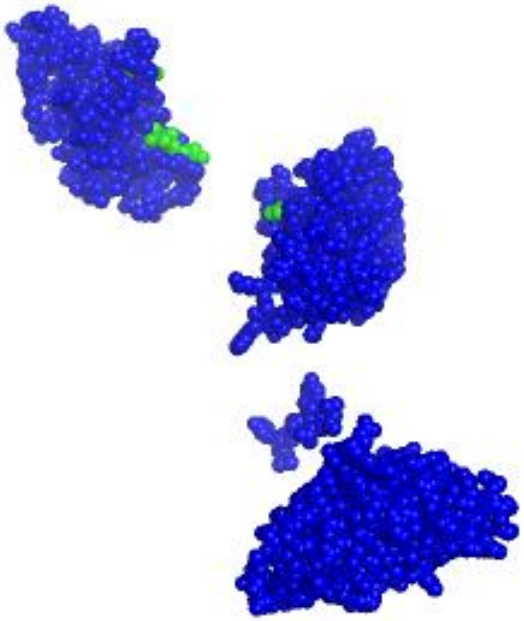
Tom Peacock

Contact hotspots

TCR alpha:TCRbeta contacts;

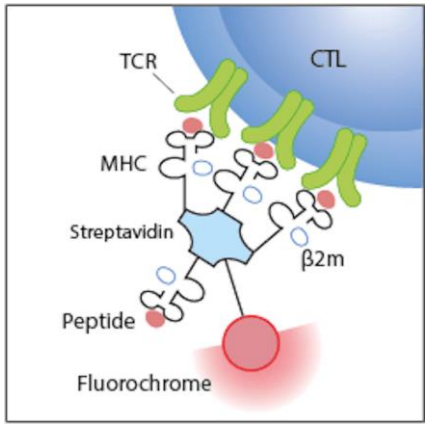
TCR:peptide

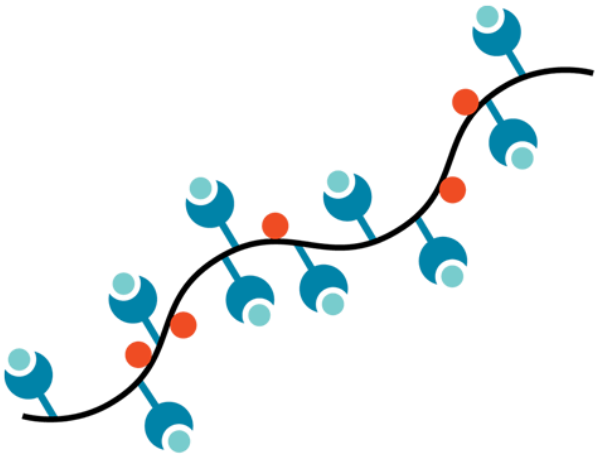
TCR : MHC

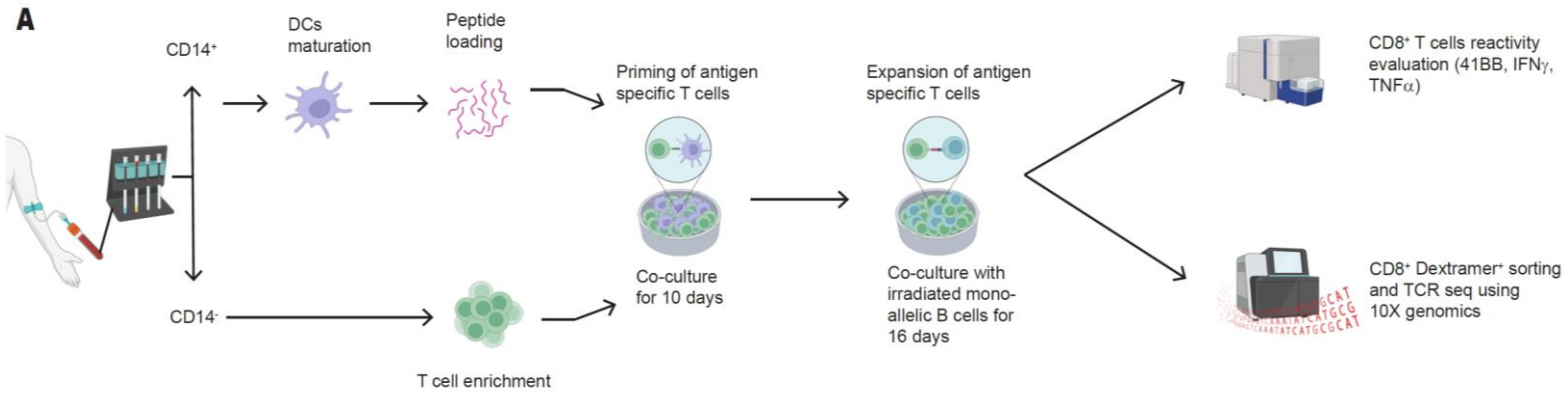


MHC:TCR:Peptide (1AO7) HLA A02 Tax

MHC, peptide and T cell identification

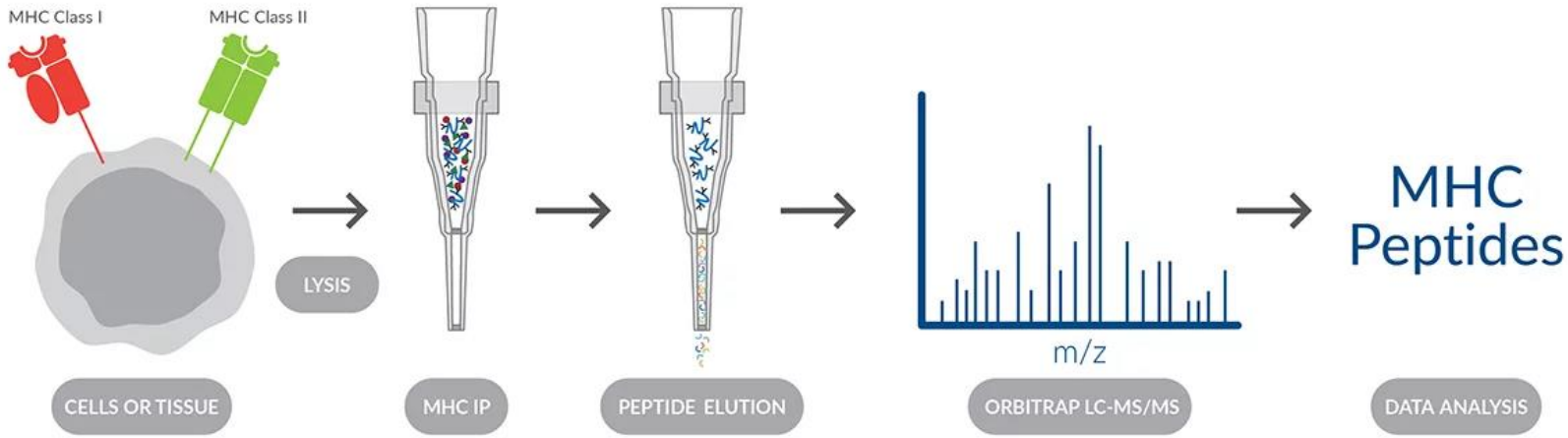






Nakandakari-Higa et al. Universal recording of immune cell interactions in vivo. Nature. 2024 Mar;627(8003):399-406. doi: 10.1038/s41586-024-07134-4. Epub 2024 Mar 6. PMID: 38448581; PMCID: PMC11078586.

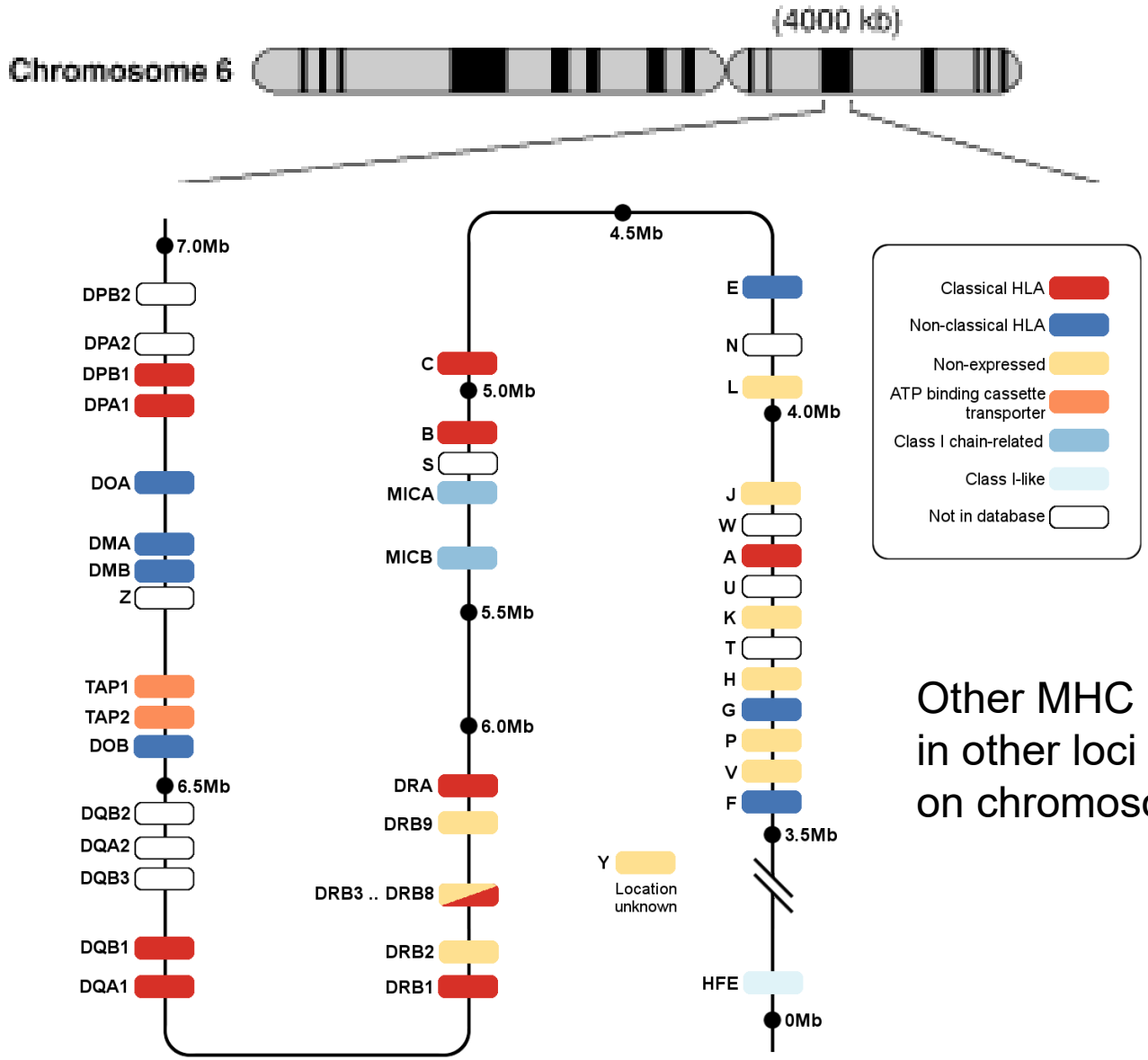
Characterising the immunopeptidome



Ely et al. Pancreatic cancer-restricted cryptic antigens are targets for T cell recognition. Science. 2025 May 8;388(6747):eadk3487. doi: 10.1126/science.adk3487.
Kalaora S, et al Identification of bacteria-derived HLA-bound peptides in melanoma. Nature. 2021 Apr;592(7852):138-143. doi: 10.1038/s41586-021-03368-8. Epub 2021 Mar 17. PMID: 33731925.
Yang et al . Autoimmunity-associated T cell receptors recognize HLA-B*27-bound peptides. Nature. 2022 Dec;612(7941):771-777. doi: 10.1038/s41586-022-05501-7. Epub 2022 Dec 7. PMID: 36477533.
Huang et alTumour circular RNAs elicit anti-tumour immunity by encoding cryptic peptides. Nature. 2024 Jan;625(7995):593-602. doi: 10.1038/s41586-023-06834-7. Epub 2023 Dec 13. PMID: 38093017.

Genetics of the MHC (HLA in humans) complex

- Classical Class I and II peptide binders
- Other (non-classical) class I and II
- MHC as NK inhibitors
- Several clusters of other genes in region, some with immunological function



Other MHC paralogs in other loci (e.g. CD1 on chromosome 1)

For further details see : HLA dictionary

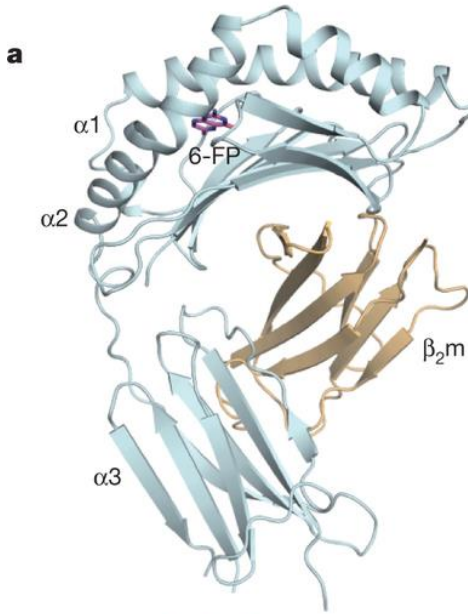
Non-classical MHC

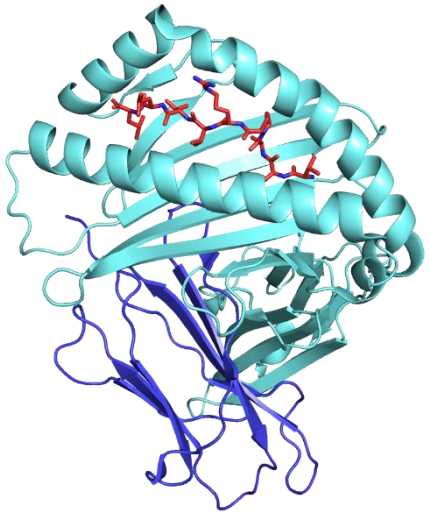
Luoma AM, et al. Crystal structure of Vδ1 T cell receptor in complex with CD1d-sulfatide shows MHC-like recognition of a self-lipid by human γδ T cells. Immunity. 2013 Dec 12;39(6):1032-

Kjer-Nielsen et al. (2012) MR1 presents microbial vitamin B metabolites to MAIT cells. Nature 491 :717-23.

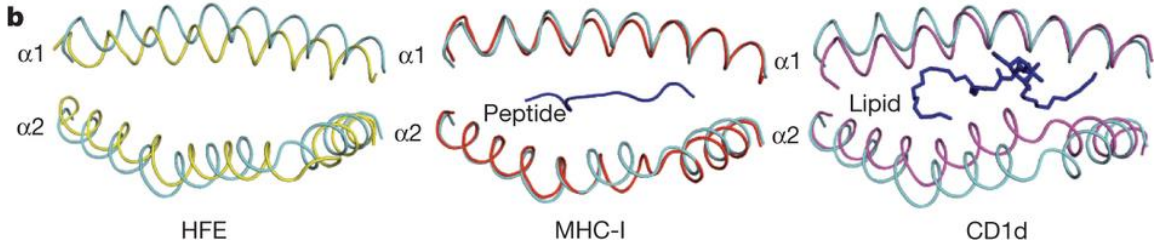
D'Souza et al. Casting a wider net: Immunosurveillance by nonclassical MHC molecules. PLoS Pathog. 2019 Feb 21;15(2):e1007567. doi: 10.1371/journal.ppat.1007567. PMID: 30789961; PMCID: PMC6383864.

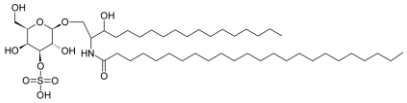
MR1
6-formyl pterin





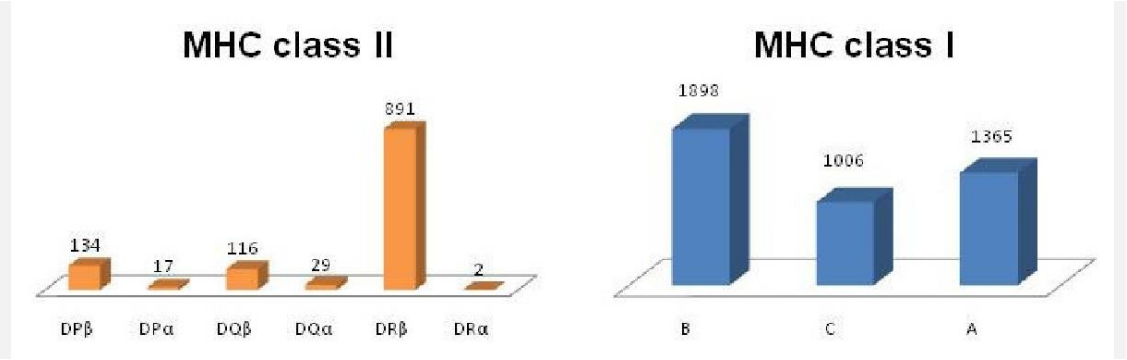
HLA-E
Non-polymorphic
NK receptor

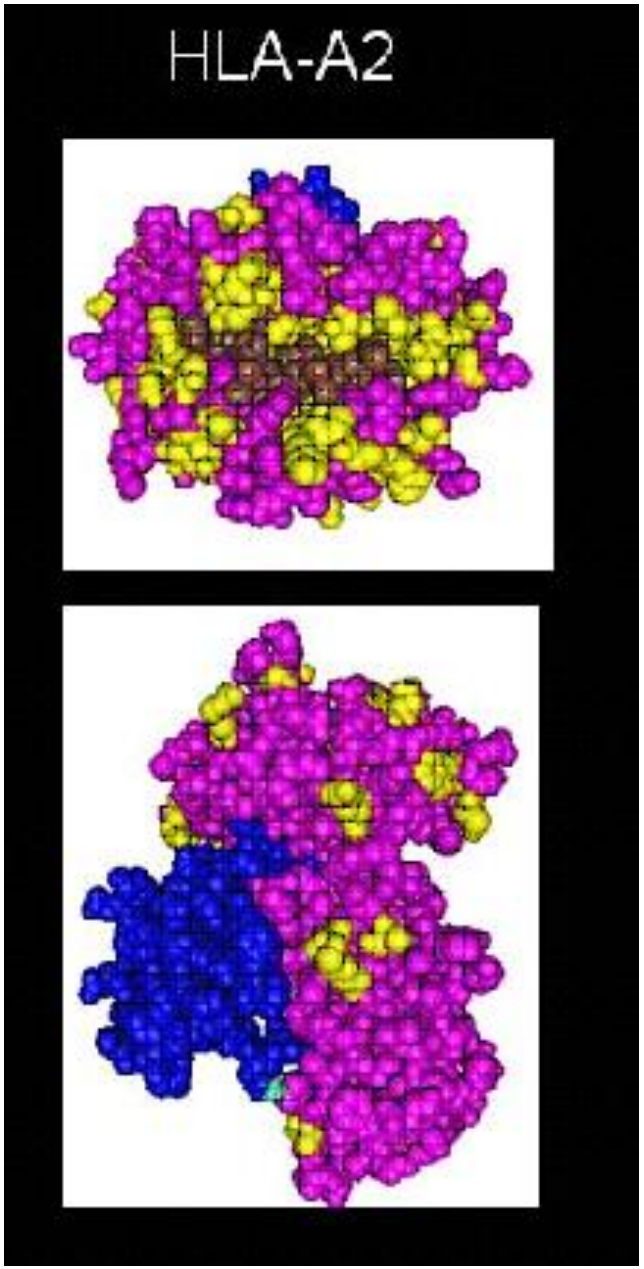




Diversity in the MHC

- Paralogs and polymorphism
- How does diversity impact recognition
- In mammals, polymorphism very patchy; absent in class I of Syrian hamster, domestic cat, low in seals; but high in desert mole rat.





How does polymorphism evolve?

Mechanisms for balanced polymorphism

- Heterozygote advantage
- Frequency dependent selection
- Sex

- MHC polymorphism and infection
- MHC polymorphism and autoimmunity
- MHC polymorphism and cancer

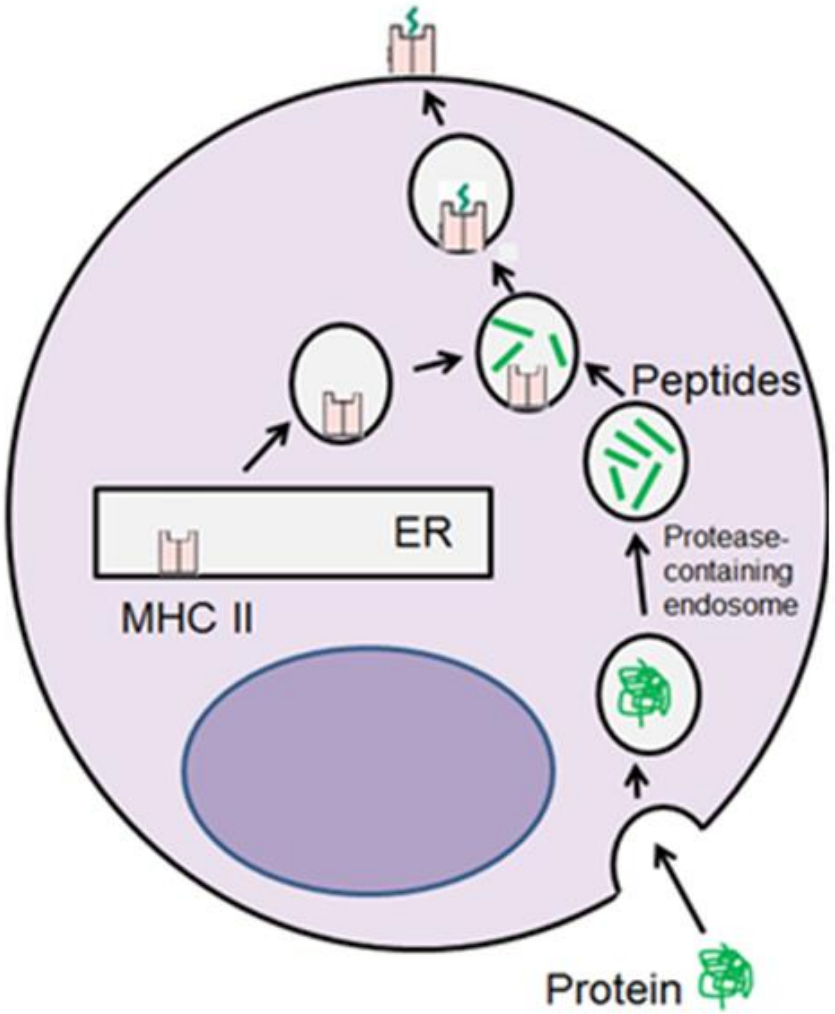
Bolnick DI, Stutz WE. Frequency dependence limits divergent evolution by favouring rare immigrants over residents. Nature. 2017 Jun 8;546(7657):285-288.
Chowell Det al. Patient HLA class I genotype influences cancer response to checkpoint blockade immunotherapy. Science. 2018 Feb 2;359(6375):582-587.
Viard M, et al Impact of HLA class I functional divergence on HIV control. Science. 2024 Jan 19;383(6680):319-325. doi: 10.1126/science.adk0777. Epub 2024 Jan 18. PMID: 38236978; PMCID: PMC11395297.

II Antigen processing

- The two pathway model
- Exogenous and endogenous

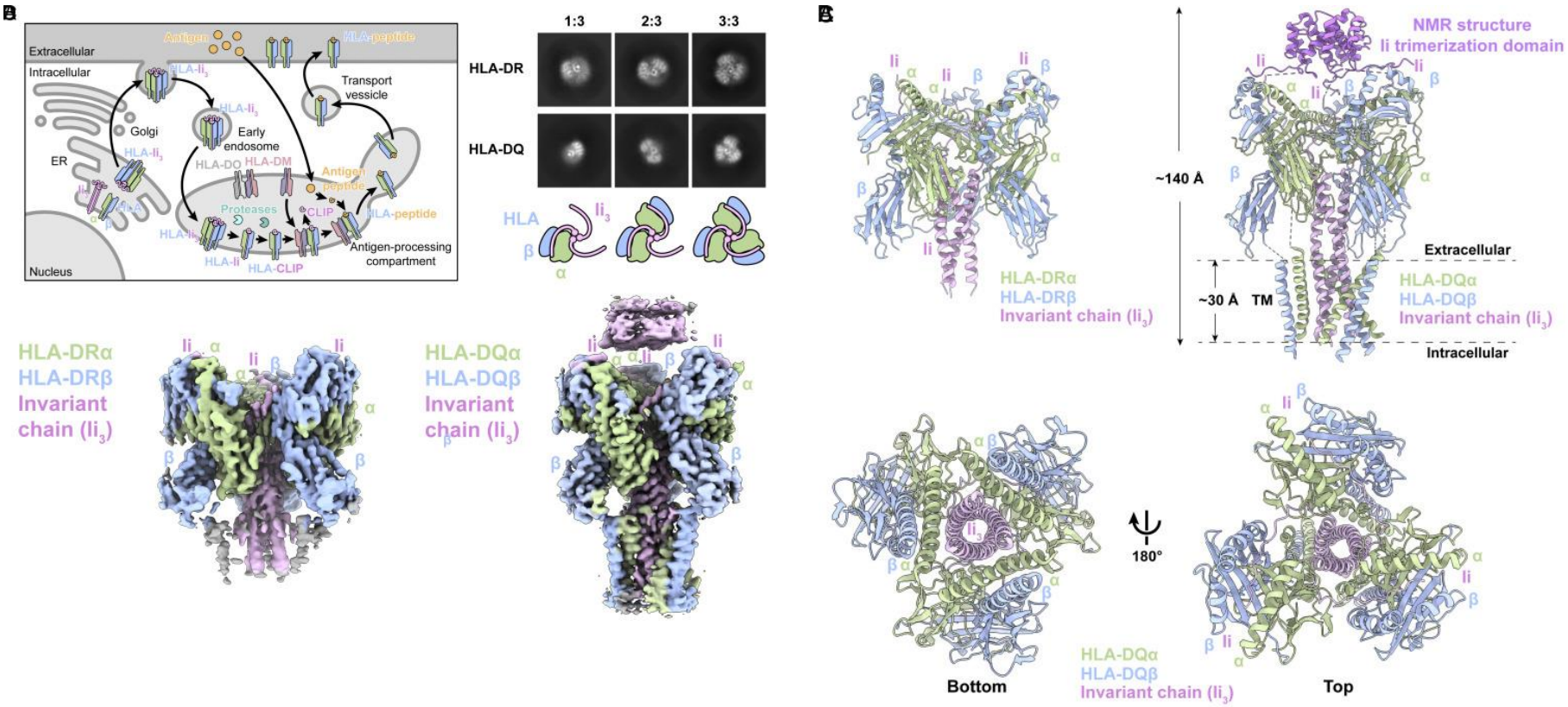
Exogenous processing

MHC class II + peptide
= Target for CD4⁺ T-cells

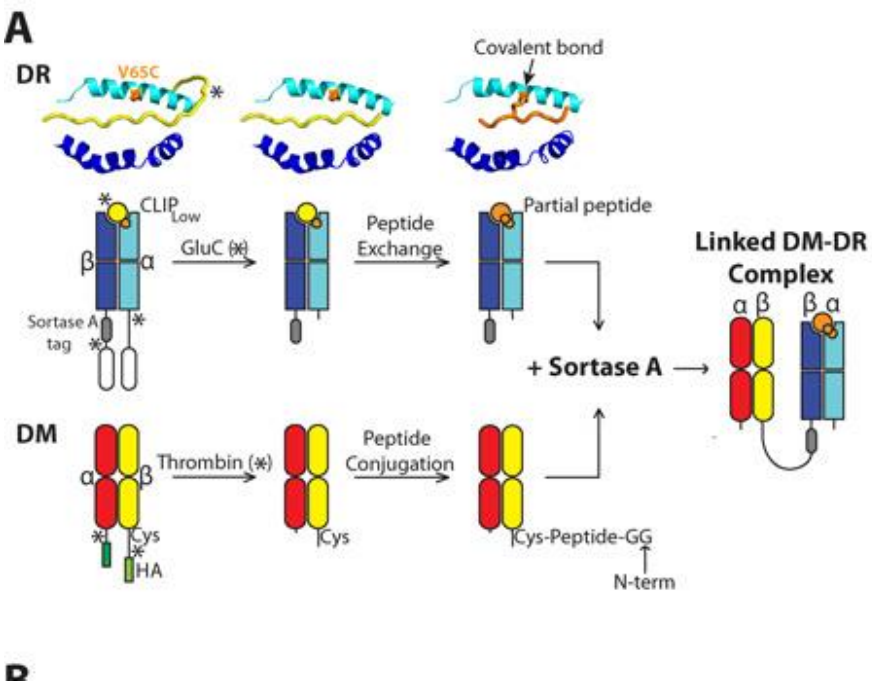


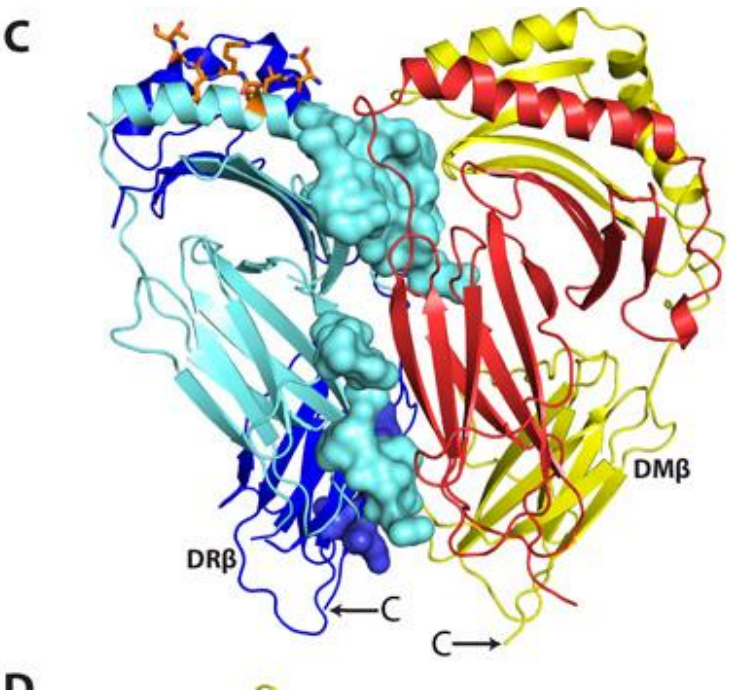
Exogenous processing

- Uptake – Pattern recognition receptors
- The loading compartment; meeting MHC class II , pH and restricted degradation
- Invariant chain
- DM and DO
- processing and the repertoire.



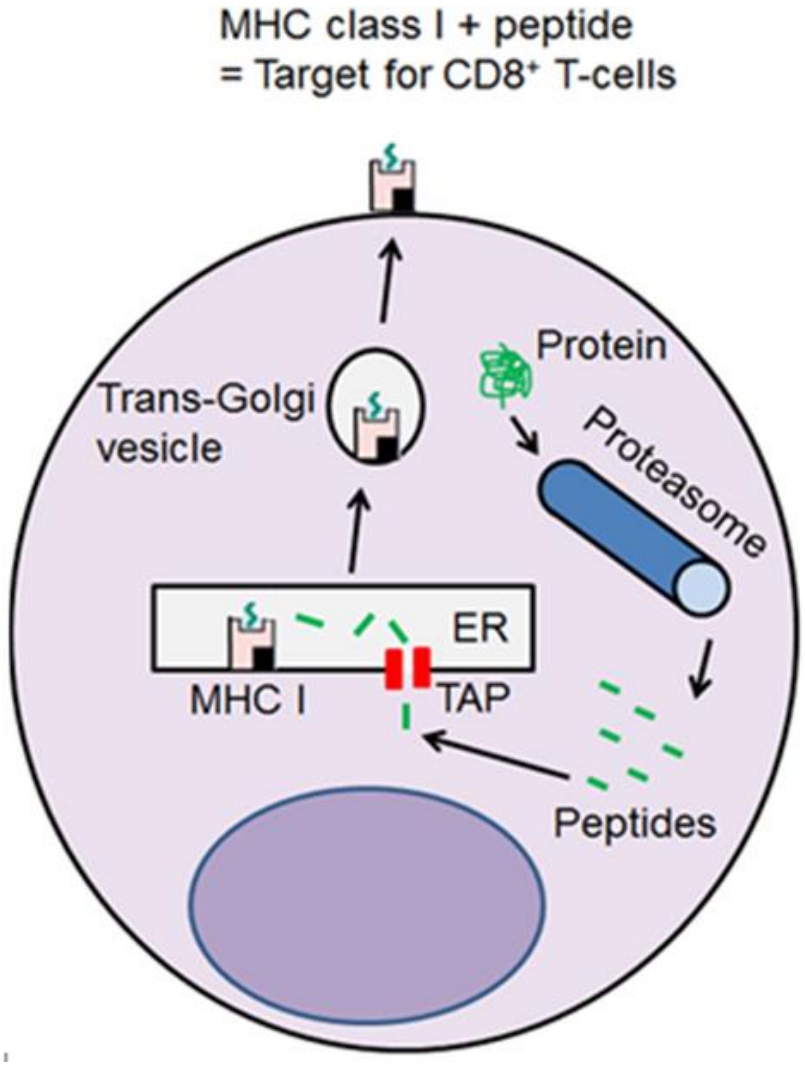
Wang N, Waghray D, Caveney NA, Jude KM, Garcia KC. Structural insights into human MHC-II association with invariant chain. Proc Natl Acad Sci U S A. 2024 May 7;121(19):e2403031121. doi: 10.1073/pnas.2403031121. Epub 2024 Apr 30. PMID: 38687785; PMCID: PMC11087810.





Pos et al. (2012) Crystal Structure of the HLA-DM - HLA-DR1 Complex Defines Mechanisms for Rapid Peptide Selection. Cell 151, 1557

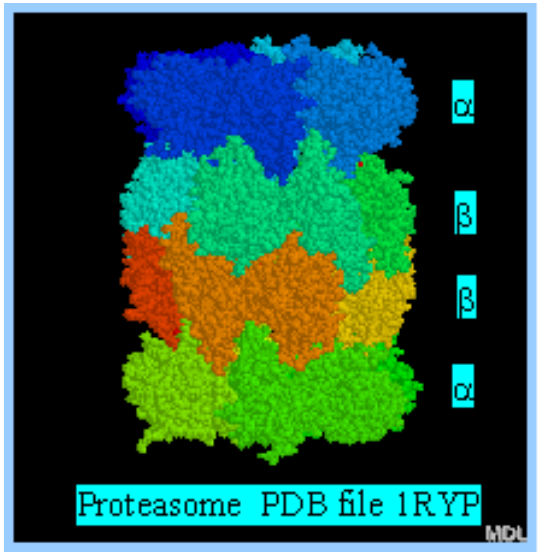
Endogenous processing

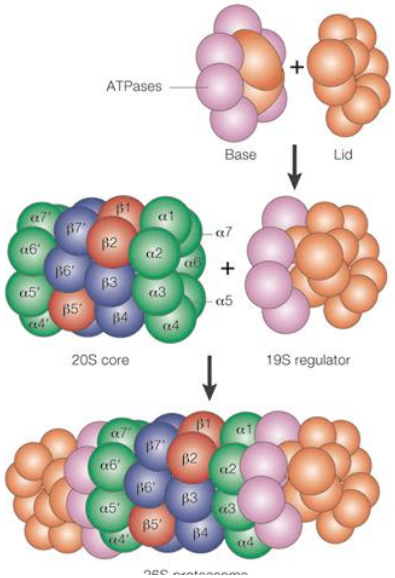


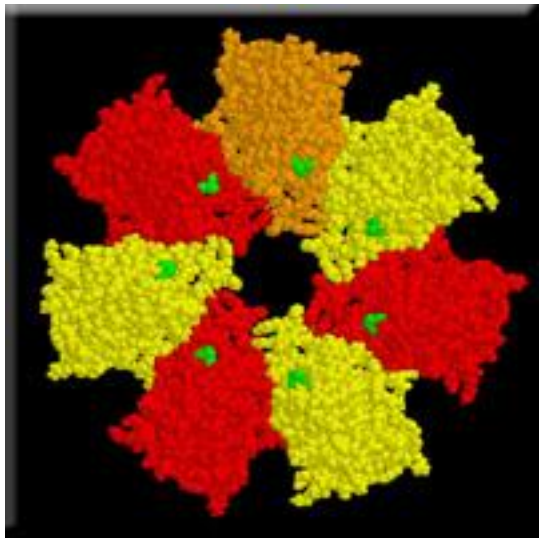
Endogenous processing

- Cytoplasmic degradation : the immune proteasome and ubiquitin.
- Crossing the ER : TAP genes and transporters
- The peptide-loading complex
- Trimming and ERAP
- Dark matter !

Cytoplasmic degradation : the proteasome



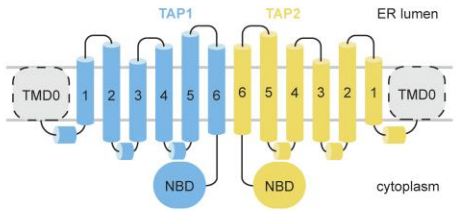




http://www.hhmi.org/biointeractive/proteasome

Liepe et al. (2017) a large fraction of HLA class I ligands are proteasome-generated spliced peptides. Science 354, 354 – 357

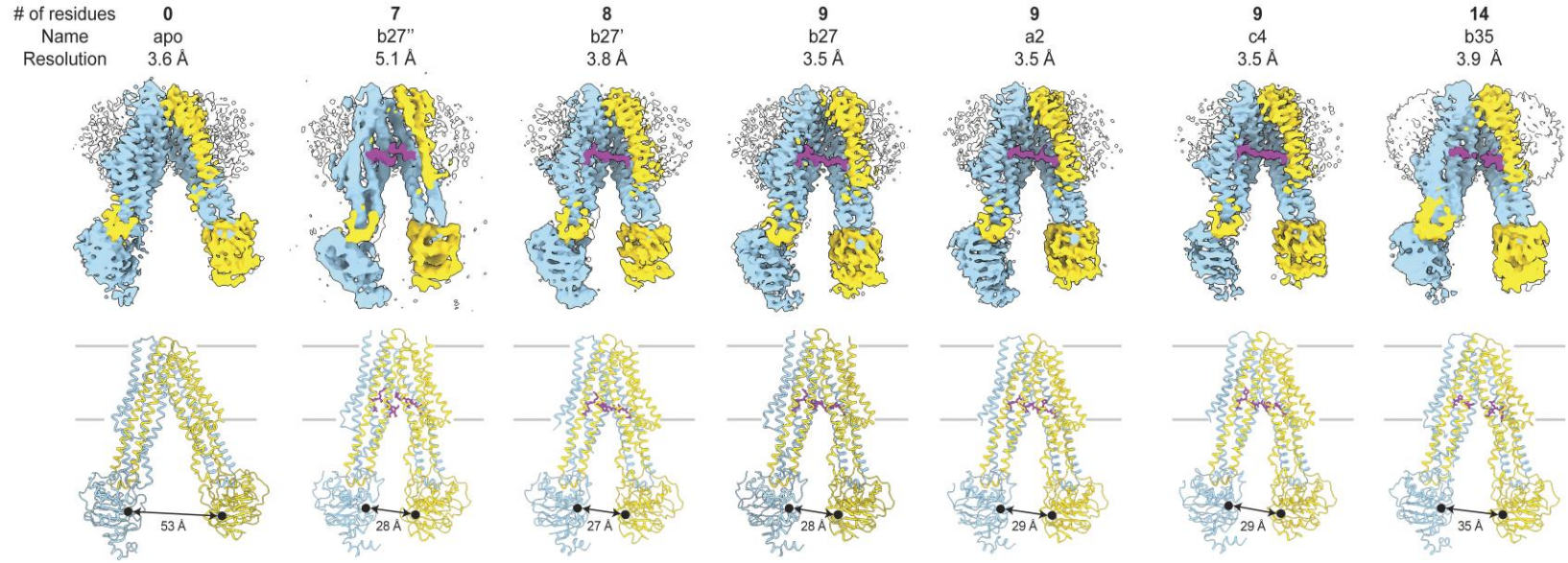
A



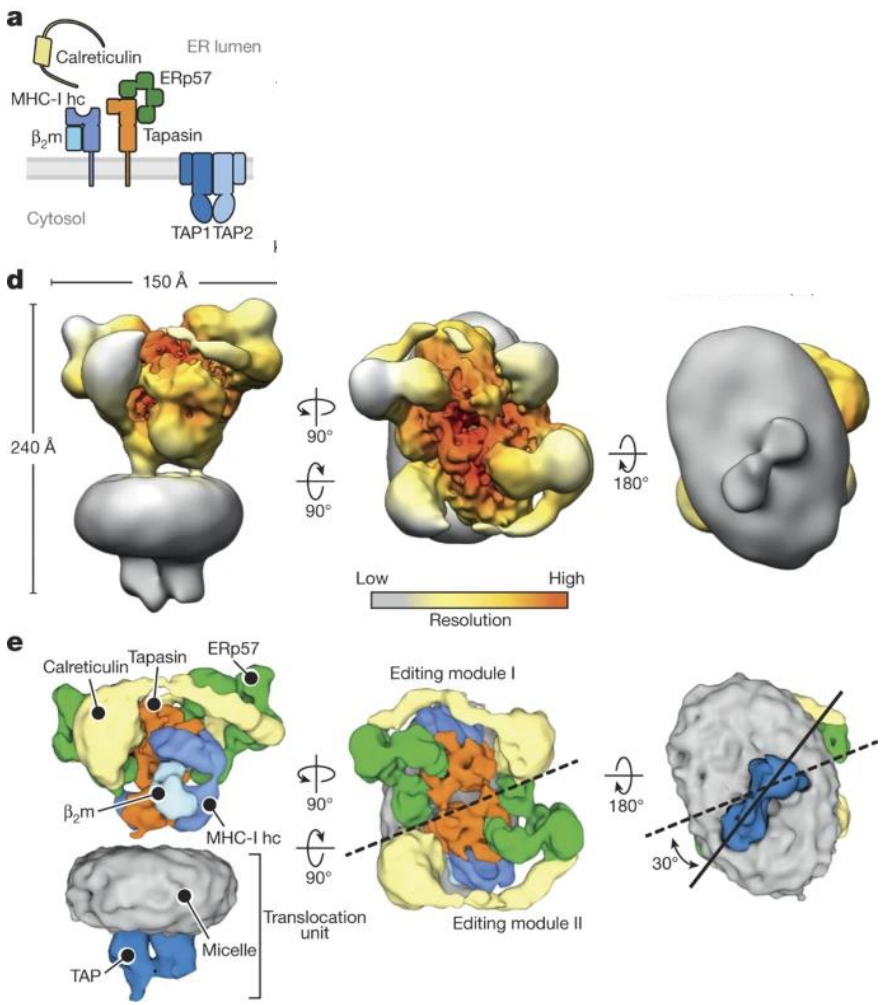
D

# of residues	Peptide	Sequence
7	b27''	RRY--STEL
8	b27'	RRYQ-STEL
9	b27	RRYQKSTEL
9	a2	ILKEPVHGV
9	c4	QYDDAVYKL
14	b35	LPAVVGLSPGEQEY

E



Lee J, Oldham ML, Manon V, Chen J. Principles of peptide selection by the transporter associated with antigen processing. Proc Natl Acad Sci U S A. 2024 Jun 4;121(23):e2320879121. doi: 10.1073/pnas.2320879121. Epub 2024 May 28. PMID: 38805290; PMCID: PMC11161800.

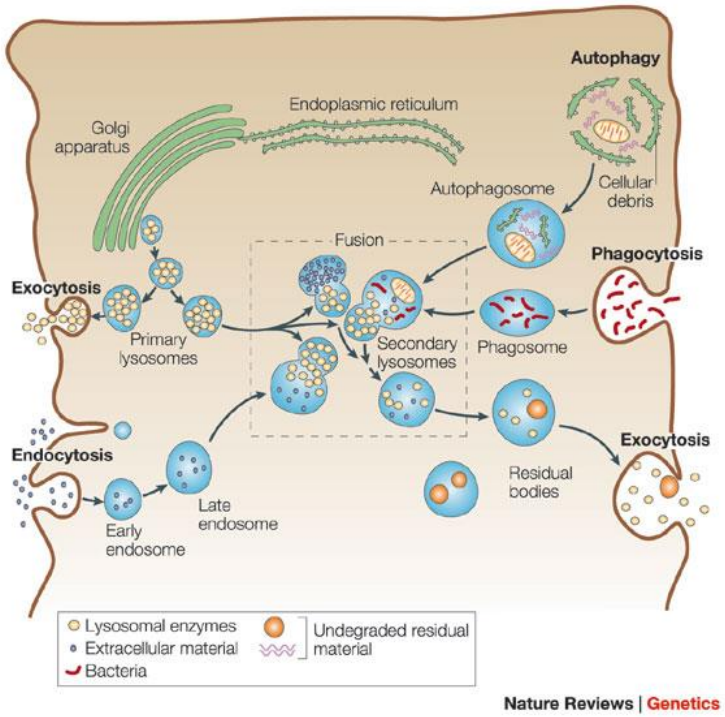


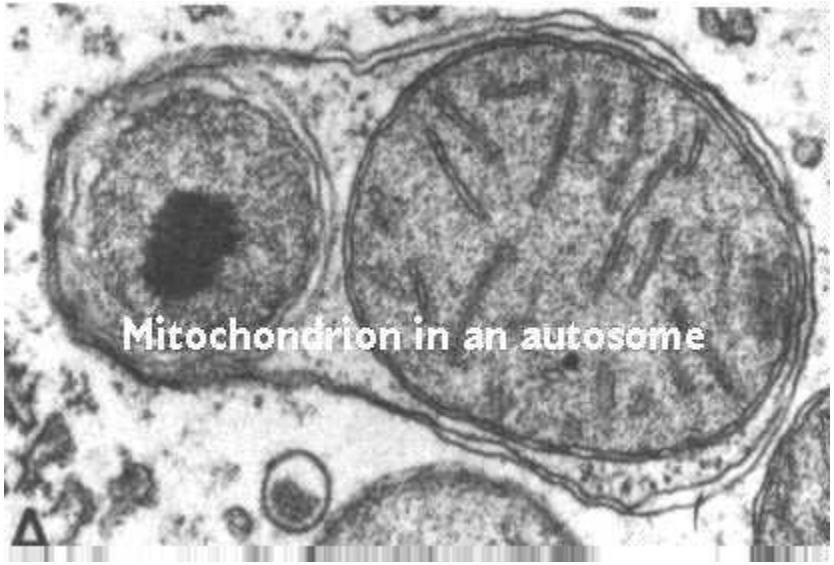
Blees A, Structure of the human MHC-I peptide-loading complex. Nature. 2017 Nov 23;551(7681):525-528.
Thomas C, Tampé R. Structure of the TAPBPR-MHC I complex defines the mechanism of peptide loading and editing. Science. 2017 Nov 24;358(6366):1060-1064.

Exceptions to the two pathway rule

- Cross presentation via class I; pH and NADPH; ERAD
- Autophagy via class II; ATG genes; ubiquitin

Rodríguez-Silvestre et al. . Perforin-2 is a pore-forming effector of endocytic escape in cross-presenting dendritic cells. Science. 2023 Jun 23;380(6651):1258-1265.


Autophagy
Endoplasmic reticulum
Golgi apparatus
Autophagosome
Cellular debris
Fusion
Phagocytosis
Exocytosis
Primary lysosomes
Secondary lysosomes
Phagosome
Endocytosis
Exocytosis
Residual bodies
Late endosome
Early endosome
Lysosomal enzymes
Extracellular material
Bacteria
Undegraded residual material
Nature Reviews | Genetics


Mitochondrion in an autosome

Regulating antigen processing

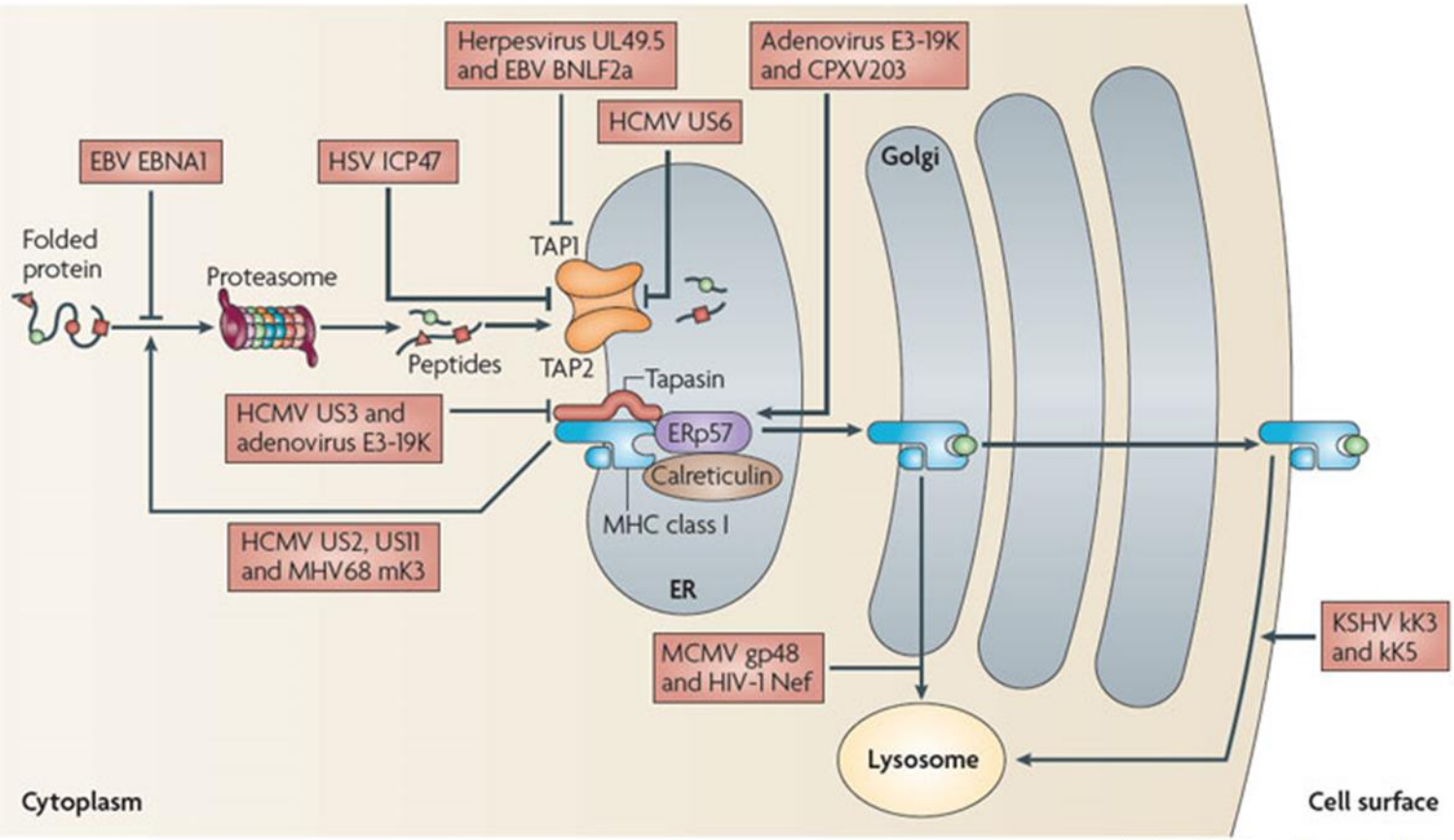
- Professional and non-professional presenting cells
- Dendritic cell subsets
- Interferon and the processing and presentation locus; IL10
- Dendritic cell maturation and the control of degradation

Tang H, et al Rescuing dendritic cell interstitial motility sustains antitumour immunity. Nature. 2025 Sep;645(8079):244-253. doi: 10.1038/s41586-025-09202-9. Epub 2025 Jun 25. PMID: 40562926.

Antigen processing and pathology

- Viral evasion
- Tumour evasion; peptide/MHC and neoantigens
- Autoimmunity and aberrant processing; modified self

Verweij et al. (2015) Viral inhibition of the transporter associated with antigen processing (TAP): a striking example of functional convergent evolution. PLoS Pathog. 16;11(4):e1004743
Puttick et al. MHC Hammer reveals genetic and non-genetic HLA disruption in cancer evolution. Nat Genet. 2024 Oct;56(10):2121-2131. doi: 10.1038/s41588-024-01883-8. Epub 2024 Oct 2. PMID: 39358601; PMCID: PMC11525181.
Javitt et al. The proteasome regulator PSME4 modulates proteasome activity and antigen diversity to abrogate antitumor immunity in NSCLC. Nat Cancer. 2023 May;4(5):629-647.
Huber F, Bassani-Sternberg M. Defects in antigen processing and presentation: mechanisms, immune evasion and implications for cancer vaccine development. Nat Rev Immunol. 2025 Aug 8. doi: 10.1038/s41577-025-01208-8. Epub ahead of print. PMID: 40781552.
Delong et al. (2016) Pathogenic CD4 T cells in type 1 diabetes recognize epitopes formed by peptide fusion. Science 351, 6274
38448581



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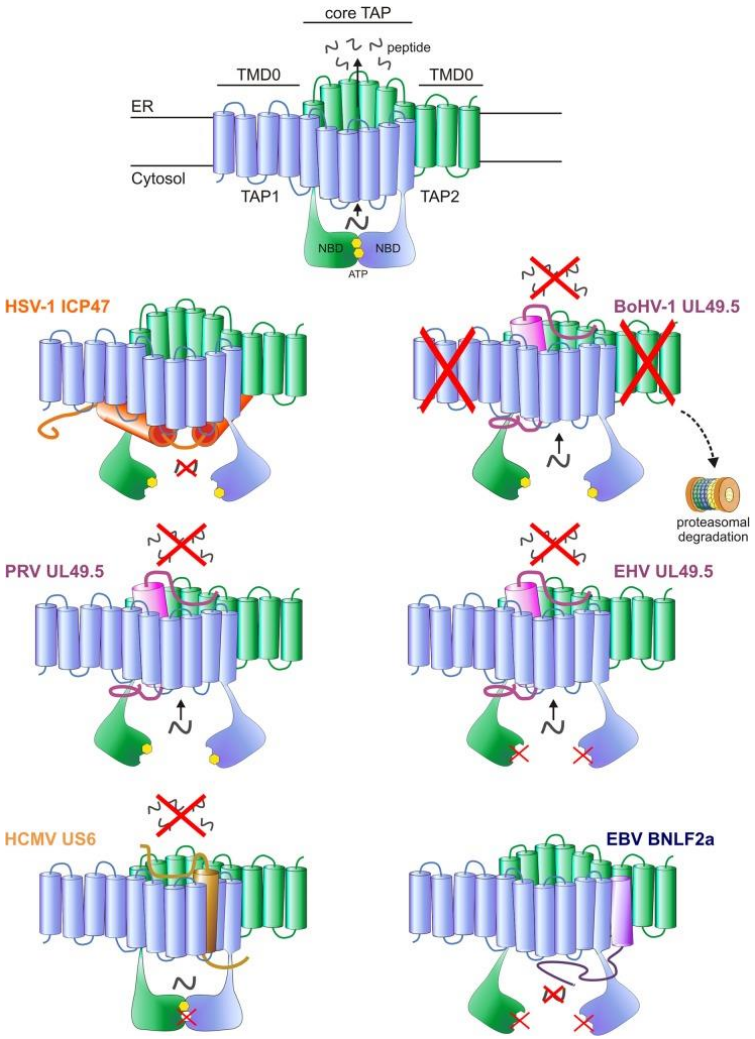


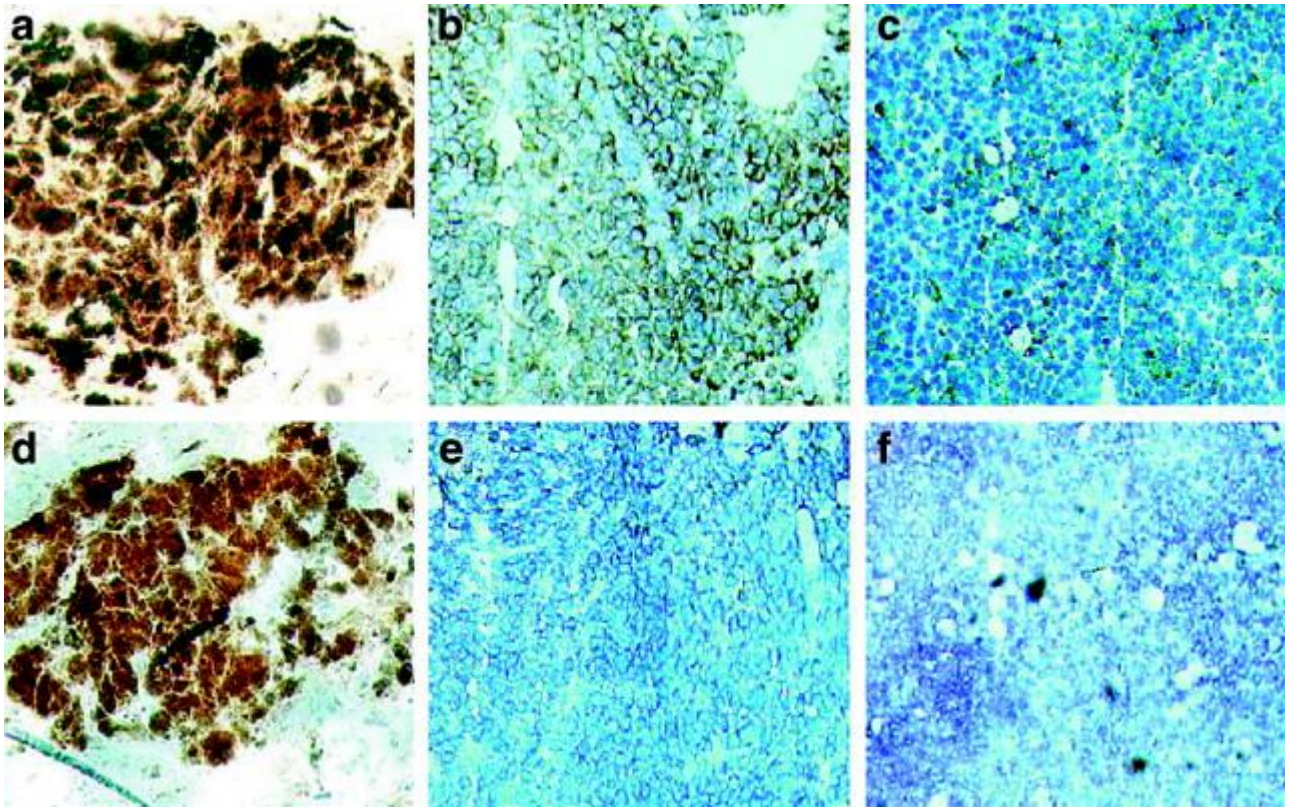
Verweij et al. (2015) Viral inhibition of the transporter associated with antigen processing (TAP): a striking example of functional convergent evolution. PLoS Pathog. 16;11(4):e1004743

Selective allele loss in metastatic melanoma

Patient 1 Patient 2 Patient 3

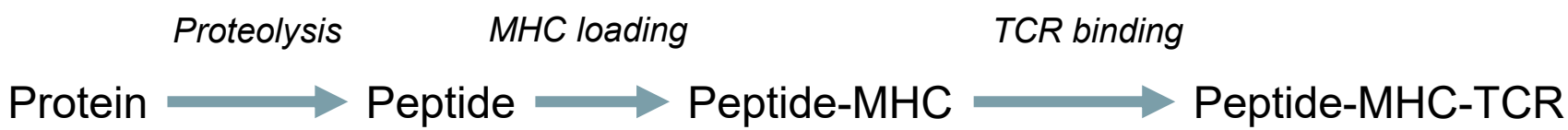
Pan-MHC

Allele specific HLA-A



Future horizons

Computational prediction of antigen processing and presentation



Reynisson et al. NetMHCpan-4.1 and NetMHCIIpan-4.0: improved predictions of MHC antigen presentation by concurrent motif deconvolution and integration of MS MHC eluted ligand data. Nucleic Acids Res. 2020 Jul 2;48(W1):W449-W454. doi: 10.1093/nar/gkaa379.

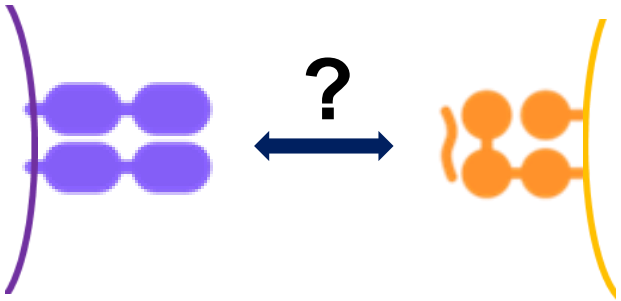
Lu et al. Deep learning-based prediction of the T cell receptor-antigen binding specificity. Nat Mach Intell. 2021 Oct;3(10):864-875. doi: 10.1038/s42256-021-00383-2. Epub 2021 Sep 23. PMID: 36003885; PMCID: PMC9396750.

Wang M, Fan W, Wu T, Li M. TPepRet: a deep learning model for characterizing T-cell receptors-antigen binding patterns. Bioinformatics. 2024 Dec 26;41(1):btaf022. doi: 10.1093/bioinformatics/btaf022.

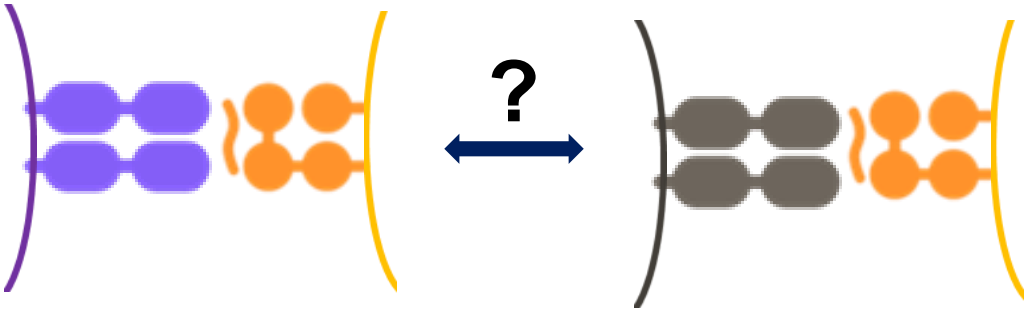
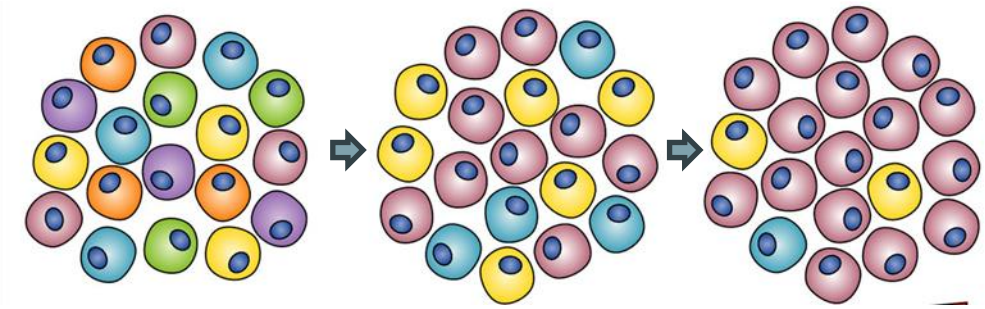
Liu B, Greenwood NF, Bonzanini JE, Motmaen A, Meyerberg J, Dao T, Xiang X, Ault R, Sharp J, Wang C, Visani GM, Vafeados DK, Roullier N, Nourmohammad A, Scheinberg DA, Garcia KC, Baker D. Design of high-specificity binders for peptide-MHC-I complexes. Science. 2025 Jul 24;389(6758):386-391. doi: 10.1126/science.adv0185. Epub 2025 Jul 24. PMID: 40705892.

The challenge

- Predicting peptide specificity from sequence – very hard



- Predicting peptide co-specificity from sequence – first step ?



Predicting repertoire specificity – diagnosis and tracking